Natural Language Processing Coursework Report

Detecting Patronising and Condescending Language (PCL)

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GitHub Repository:
https://github.com/GiannisKitsos/NLP-PCL-Detection/tree/main

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Contents

1 Exercise 1: Critical Paper Review **2**
1.1 Q1. Primary Contributions of This Work (2 Marks) . . . 2
1.2 Q2. Technical Strengths (2 Marks) . . . 2
1.3 Q3. Key Weaknesses and Insufficient Evidence (2 Marks) . . . 3

2 Exercise 2: Exploratory Data Analysis (EDA) **4**
2.1 EDA Technique 1: Class Distribution and Imbalance Analysis (3 Marks) . . . 4
2.1.1 Visual/Tabular Evidence . . . 4
2.1.2 Analysis . . . 4
2.1.3 Impact Statement . . . 5
2.2 EDA Technique 2: Sequence Length Distribution and Linguistic Patterns (3 Marks) 5
2.2.1 Visual/Tabular Evidence . . . 6
2.2.2 Analysis . . . 7
2.2.3 Impact Statement . . . 7

3 Exercise 3: Proposed Approach **8**
3.1 Proposed Approach (2 Marks) . . . 8
3.1.1 1. Upgrade to RoBERTa-large . . . 8
3.1.2 2. Keyword Prefix as Auxiliary Input . . . 8
3.1.3 3. Focal Loss for Class Imbalance . . . 8
3.1.4 Training Configuration . . . 8
3.2 Rationale and Expected Outcome (2 Marks) . . . 9
3.2.1 Rationale . . . 9
3.2.2 Expected Outcome . . . 9

4 Exercise 5.2: Local Evaluation **10**
4.1 Error Analysis (2.5 Marks) . . . 10
4.1.1 Methodology . . . 10
4.1.2 Error Pattern 1: False Positives — Advocacy Confused with PCL . . . 10
4.1.3 Error Pattern 2: False Negatives — Subtle PCL Missed . . . 10
4.1.4 Keyword Error Distribution . . . 11
4.1.5 Summary of Error Analysis Findings . . . 11
4.2 Additional Local Evaluation: Ablation Study and Precision-Recall Analysis (2.5 Marks) . . . 11
4.2.1 Ablation Study . . . 11
4.2.2 Precision-Recall Curve . . . 12
4.2.3 Confusion Matrix . . . 13

1 Exercise 1: Critical Paper Review

1.1 Q1. Primary Contributions of This Work (2 Marks)

The paper "Don't Patronize Me!" introduces three primary contributions:

1. **A Novel Annotated Dataset**: The authors present the first large-scale dataset specifically designed for identifying Patronizing and Condescending Language (PCL) toward vulnerable communities. The dataset contains 10,637 paragraphs extracted from news stories across 20 English-speaking countries, covering various vulnerable groups (disabled, homeless, immigrants, refugees, poor families, and women). This addresses a significant gap in NLP research, as previous work on harmful language focused primarily on explicit forms like hate speech rather than subtle, well-intentioned yet condescending discourse.

2. **A Two-Level Taxonomy of PCL Categories**: The paper introduces a comprehensive categorization system with seven subcategories grouped into three higher-level categories:

   - **The Saviour** (Unbalanced power relations, Shallow solution)
   - **The Expert** (Presupposition, Authority voice)
   - **The Poet** (Metaphor, Compassion, The poorer the merrier)

   This taxonomy provides a framework for understanding the linguistic manifestations of PCL and enables fine-grained analysis.

3. **Baseline Models and Experimental Analysis**: The authors provide comprehensive baseline results using various approaches (SVM with Word2Vec/BoW, BiLSTM, and fine-tuned BERT models [1, 3]) on two tasks: (1) binary classification to detect PCL presence, and (2) multi-label classification to categorize PCL types. Their analysis reveals that while modern language models (particularly RoBERTa) achieve reasonable performance on certain categories, detecting PCL remains challenging, especially for categories requiring world knowledge (e.g., Metaphor, Presupposition, Shallow solution).

1.2 Q2. Technical Strengths (2 Marks)

The paper demonstrates several technical strengths that justify its publication:

1. **Rigorous Annotation Methodology**: The authors employed three expert annotators with relevant backgrounds in communication, media, and data science. They used a two-step annotation process: first identifying PCL at the paragraph level using a 5-point scale to capture nuanced disagreements, then annotating specific text spans with category labels. The inclusion of inter-annotator agreement (IAA) scores (ranging from 41% overall to 61% when excluding borderline cases) demonstrates transparency about the subjective nature of the task. The use of a third annotator as a referee for total disagreements (0 vs 2) is methodologically sound.

2. **Comprehensive Experimental Setup**: The paper provides extensive baseline experiments using diverse approaches, from traditional machine learning (SVM with different representations) to modern deep learning (BiLSTM) and state-of-the-art language models (multiple BERT variants [1, 3]). The use of 10-fold cross-validation ensures robust evaluation. The authors evaluate both coarse-grained (binary PCL detection) and fine-grained (category classification) tasks, providing insights at multiple levels of granularity.

3. **Thoughtful Error Analysis and Dataset Design**: The authors provide qualitative analysis of model failures (Tables 5 and 6), identifying specific challenges such as detecting

metaphors and presuppositions that require world knowledge. The dataset design is balanced across keywords and countries (Table 1), reducing potential biases. The inclusion of borderline cases (labels 1, 2, 3) acknowledges the subjective nature of PCL and provides valuable training signal for future research.

1.3 Q3. Key Weaknesses and Insufficient Evidence (2 Marks)

Despite its contributions, the paper has several weaknesses:

1. **Limited Discussion of Dataset Biases and Generalization**: While the authors extract data from 20 countries to ensure diversity, they do not adequately address potential biases introduced by their keyword-based selection strategy. All paragraphs are retrieved using 10 specific keywords (disabled, homeless, immigrant, etc.), which may skew the dataset toward contexts where these exact terms appear. Real-world PCL may use euphemisms or indirect references that this sampling method misses. Additionally, restricting to the News on Web (NoW) corpus means the dataset reflects journalistic language patterns, which may not generalize to other domains like social media, NGO communications, or policy documents, domains mentioned as future work but not empirically validated.

2. **Insufficient Analysis of Inter-Annotator Agreement and Borderline Cases**: While the authors report moderate IAA (41% overall, 61% excluding borderline), they do not deeply investigate *why* disagreements occur or whether certain categories are inherently more subjective. The decision to combine annotator labels into a 5-point scale is pragmatic but lacks justification, for instance, why should (0+1) be treated equivalently to instances where ann3 judged a (0+2) disagreement as "not PCL"? Both receive label 1, but the underlying annotation uncertainty differs. More importantly, 867 of 1457 disagreements (59.5%) involved borderline cases, suggesting the task boundaries are unclear even for experts. The authors do not provide Cohen's Kappa for each category pair or analyze which linguistic phenomena drive disagreement.

3. **Weak Baseline Performance Without Sufficient Investigation of Causes**: The best model (RoBERTa) achieves only 70.63 F1 on Task 1 (binary classification) and struggles with several categories in Task 2, notably, all models fail on "The poorer, the merrier" (best F1: 20.5 by RoBERTa) despite this category having the highest IAA (66.72%). This counterintuitive result deserves deeper investigation. The authors note that "models struggle to detect such cases" but do not explain *why* a category that humans find "very easy to identify" is so difficult for models. Is it due to requiring complex reasoning, cultural context, or simply insufficient training examples (only 64 instances)? The paper would benefit from ablation studies isolating these factors, for example, testing whether data augmentation or few-shot learning improves performance on rare categories, or whether incorporating external knowledge bases helps with presupposition detection.

Additional Minor Weaknesses:

- The paper does not report statistical significance tests for performance differences between models.
- The choice of BERT-large performing worse than BERT-base is attributed to overfitting, but no regularization techniques or learning curves are shown to support this claim.
- The dataset's temporal range (2010-2018) is not discussed in terms of how PCL language patterns may have evolved, especially given increased awareness of inclusive language in recent years.

2 Exercise 2: Exploratory Data Analysis (EDA)

2.1 EDA Technique 1: Class Distribution and Imbalance Analysis (3 Marks)

2.1.1 Visual/Tabular Evidence

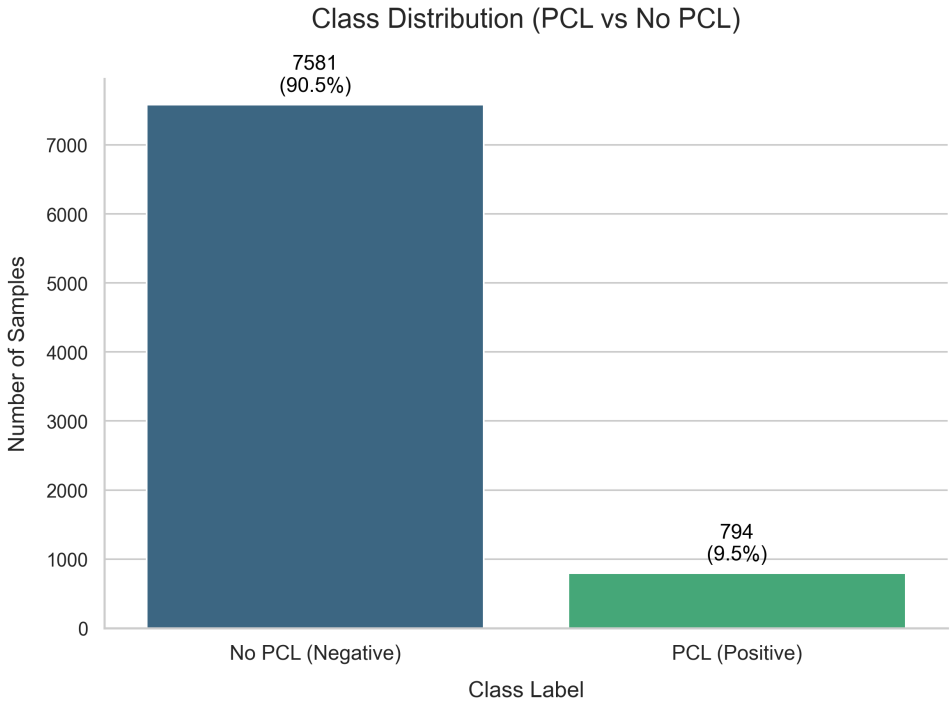

Figure 1: Class Distribution in the PCL Training Dataset

Table 1: Binary Classification Distribution Statistics

Metric	**Value**	**Percentage**
Total Training Examples	8,375	100%
No PCL (Negative Class)	7,581	90.5%
PCL (Positive Class)	794	9.5%
Class Imbalance Ratio	**9.55:1**	-

2.1.2 Analysis

The dataset exhibits severe class imbalance, with PCL examples representing only 9.5% of all paragraphs in the training set. As shown in Figure 1, the distribution is heavily skewed toward the negative class, with 7,581 non-PCL examples versus only 794 PCL examples. This approximately 1:10 ratio presents several critical challenges:

1. **Naive Baseline Problem**: A trivial classifier that always predicts "No PCL" would achieve 90.5% accuracy while completely failing to identify any patronizing content. This demonstrates why accuracy is an inappropriate metric for this task, and why the competition organizers chose F1 score of the positive class instead.

2. **Training Bias**: Standard loss functions (cross-entropy) treat all examples equally. With

this imbalance, the model receives 9.55 times more gradient updates from negative examples than positive ones, potentially causing it to:

- Learn to predict the majority class by default
- Develop overly conservative decision boundaries
- Achieve high overall accuracy but poor recall on PCL instances

3. **Limited Learning Signal**: With only 794 positive examples, the model has limited exposure to the diverse manifestations of PCL. This is particularly concerning given the paper's identification of seven distinct PCL subcategories, some categories may have fewer than 100 training examples each.

4. **Evaluation Sensitivity**: The F1 score is highly sensitive to class imbalance. A small absolute number of errors on the minority class (e.g., missing 80 of 794 PCL instances) translates to a 10% drop in recall, significantly impacting F1 score.

2.1.3 Impact Statement

This severe imbalance fundamentally shapes my modeling approach in three critical ways:

1. **Loss Function Modification**: I will replace standard cross-entropy with **Focal Loss** [2]:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \tag{1}$$

where α_t applies class weighting (setting $\alpha = 0.905$ for PCL to counteract the 90.5% negative bias) and $\gamma = 2$ down-weights easy examples. This combination ensures:

- The model continues learning from difficult PCL instances even after achieving high accuracy on easy negatives
- Misclassifying a PCL example incurs 9.55× higher penalty than misclassifying a non-PCL example
- Hard-to-classify borderline cases receive more attention during training

2. **Threshold Optimisation**: Rather than using a fixed 0.5 decision threshold, I sweep thresholds from 0.1 to 0.9 at evaluation time to find the threshold that maximises F1 on the dev set. This is important because Focal Loss suppresses output probabilities, meaning the optimal threshold for this model is typically much higher than 0.5.

2.2 EDA Technique 2: Sequence Length Distribution and Linguistic Patterns (3 Marks)

2.2.1 Visual/Tabular Evidence

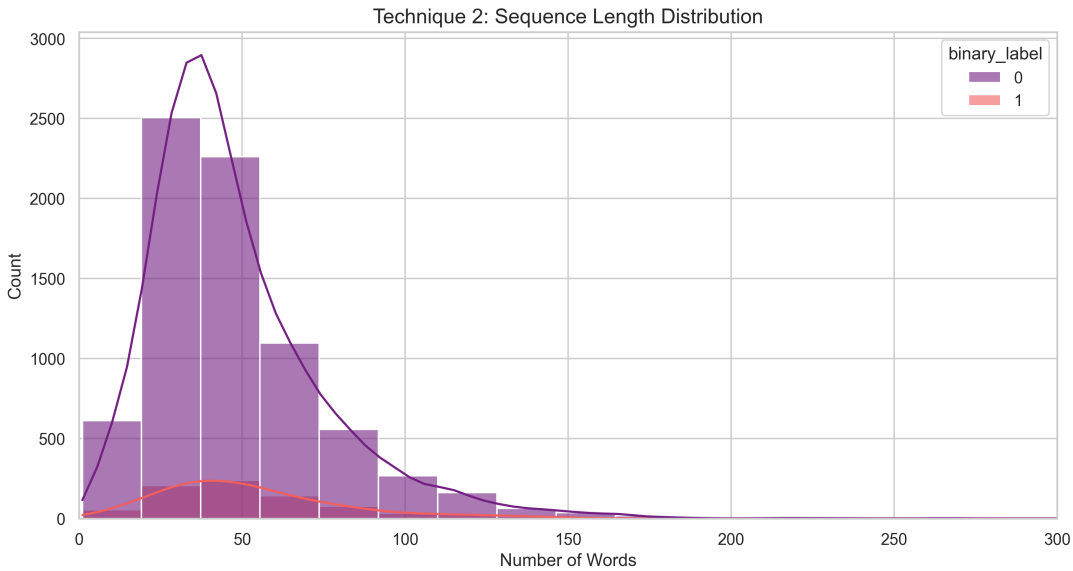

Figure 2: Distribution of paragraph lengths (in words) for PCL vs. Non-PCL examples. The overlaid density curves show that PCL paragraphs (red) are shifted toward longer lengths compared to non-PCL (purple).

Table 2: Sequence Length Statistics

Metric	**Overall**	**By Class**
Average Paragraph Length (All)	48.68 words	-
Average Length (No PCL)	-	48.17 words
Average Length (PCL)	-	53.52 words
Length Difference (PCL - No PCL)	-	**+5.35 words (+11.1%)**
Potential Noise (texts < 5 words)	11 rows (0.13%)	-

Examples of Noisy Short Texts:

- "The Filipino immigrant"
- "Volunteer helps earliest refugees"
- "UTI in pregnant women"
- "refugees"
- "Review refugee law"

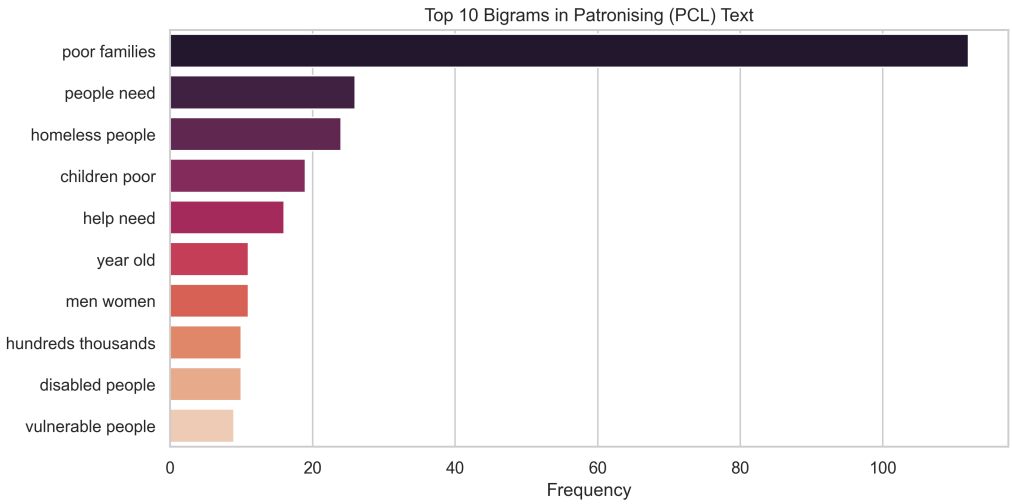

Figure 3: Top 10 bigrams (two-word phrases) in PCL-labeled text, showing frequent co-occurrence patterns.

2.2.2 Analysis

The sequence length analysis reveals that PCL paragraphs are moderately longer than non-PCL ones (+11.1% on average), consistent with the expectation that patronising language often involves more elaborate framing, emotional appeals, and qualifying language. However, the substantial overlap between the two distributions confirms that length alone is not a reliable indicator of PCL, the model must learn content-based features.

The bigram analysis (Figure 3) reveals characteristic vocabulary in PCL text: phrases such as "poor families," "homeless people," "people need," and "children poor" appear frequently, reflecting the paper's PCL taxonomy categories (Saviour, Compassion). Importantly, these same terms also appear in non-PCL text (neutral reporting about these groups), illustrating the core challenge of the task: the model must distinguish between *discussing* vulnerability and *patronising* it.

Additionally, 11 very short texts (<5 words) were identified as likely data noise (e.g., "refugees," "The Filipino immigrant"), representing incomplete paragraphs or extraction artefacts. These were removed during preprocessing.

2.2.3 Impact Statement

1. **Sequence Length Configuration**: Given that PCL paragraphs average 53.52 words and the histogram shows that most examples fall below 100 words (∼130–150 tokens), setting `max_length=256` tokens accommodates the vast majority of examples without truncation. My ablation study (Section 5.2) subsequently confirmed that 256 and 128 tokens produce near-identical F1, suggesting the relevant PCL signal is typically captured within the first 128 tokens.

2. **Keyword Prefix Strategy**: The bigram analysis showed that PCL vocabulary is closely tied to specific vulnerable groups. This motivated my keyword prefix approach ("Topic: {keyword}."), which provides the model with explicit context about which vulnerable group is being discussed, helping it distinguish patronising from neutral reporting about the same group.

3. **Text Preprocessing**: The identification of HTML artefacts, URLs, and noisy short texts motivated the preprocessing pipeline: HTML entity unescaping, URL removal, and whitespace normalisation applied before tokenisation.

3 Exercise 3: Proposed Approach

3.1 Proposed Approach (2 Marks)

My proposed approach fine-tunes **RoBERTa-large** for binary PCL classification, incorporating three key modifications over the SemEval baseline (RoBERTa-base with standard cross-entropy):

3.1.1 1. Upgrade to RoBERTa-large

I replace the baseline `roberta-base` (125M parameters) with `roberta-large` [3] (355M parameters). RoBERTa-large has deeper representations and greater capacity to capture the subtle linguistic nuances of PCL, presuppositions, metaphors, and power dynamics, that the original paper identified as the primary failure modes of smaller models.

3.1.2 2. Keyword Prefix as Auxiliary Input

Unlike Perez-Almendros et al. [5], who used topic keywords (e.g., "homeless," "refugee") only during data collection, I incorporate the keyword directly into the model input:

```
text_input = f"Topic: {keyword}. {text}"
```

This gives the model explicit knowledge of which vulnerable group is being discussed, helping it learn group-specific patterns of PCL and distinguish patronising from neutral reporting about the same group (e.g., a news article *about* homeless people vs. an article that *patronises* homeless people).

3.1.3 3. Focal Loss for Class Imbalance

I replace standard cross-entropy with Focal Loss [2] to address the severe 9.55:1 class imbalance:

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \tag{2}$$

where $\alpha_t = [1 - \alpha, \alpha]$ with $\alpha = 1 - \text{pos_ratio} \approx 0.905$, and $\gamma = 2$. The α weighting penalises false negatives proportionally to the class imbalance, while the $(1 - p_t)^{\gamma}$ term down-weights easy examples (high-confidence correct predictions) and focuses training on hard-to-classify instances.

3.1.4 Training Configuration

Table 3: Training Hyperparameters

Parameter	Value
Model	roberta-large
Epochs	6 (early stopping, patience=2)
Learning rate	8×10^{-6}
Batch size	8 (effective: 32 with gradient accumulation steps=4)
Weight decay	0.01 [4]
Warmup ratio	0.06
LR scheduler	Cosine annealing
Loss function	Focal Loss ($\alpha = 1 - \text{pos_ratio}$, $\gamma = 2$)
fp16	True
max_length	256 tokens

Text Preprocessing: HTML entity unescaping, URL removal, and whitespace normalisation applied to all splits before tokenisation. The standard RoBERTa BPE tokeniser is used.

Threshold Optimisation: At inference time, classification threshold is selected by sweeping values from 0.1 to 0.9 (step 0.01) on the dev set and choosing the threshold that maximises F1.

3.2 Rationale and Expected Outcome (2 Marks)

3.2.1 Rationale

1. **Why RoBERTa-large over RoBERTa-base?** The original paper showed RoBERTa-base outperformed all other models including BERT-large, which overfitted. RoBERTa-large has double the layers (24 vs 12) and attention heads (16 vs 12), giving it greater representational capacity for nuanced tasks like presupposition detection and metaphor identification. The key concern is overfitting on this relatively small dataset (∼8,000 training examples), which is mitigated by a low learning rate (8×10^{-6}), cosine scheduling with warmup, and early stopping.

2. **Why the keyword prefix?** The EDA and bigram analysis showed that PCL vocabulary is closely tied to specific vulnerable groups. By prepending "Topic: {keyword}.", the model receives explicit signal about the discourse context. Crucially, this addresses the model's main failure mode: confusing neutral reporting *about* a group (e.g., "A clinic helps a Syrian refugee") with language that is patronising *toward* that group. The paper did not explore this approach, making it a novel contribution.

3. **Why Focal Loss?** Standard cross-entropy with a 9.55:1 imbalance causes the model to optimise for the majority class. Focal Loss's α-weighting directly counteracts this by assigning proportionally higher loss to minority class errors. The $(1-p_t)^\gamma$ term addresses a second problem: with cross-entropy, once the model becomes confident on easy negatives, further training produces diminishing gradients. Focal Loss ensures difficult PCL examples continue generating meaningful updates throughout training.

4. **Why cosine scheduling with warmup?** RoBERTa-large has 355M parameters. Without warmup, the large learning rate in early steps can destabilise the pretrained weights. Cosine decay ensures a smooth reduction in learning rate across the training run, preventing abrupt changes that could cause catastrophic forgetting.

3.2.2 Expected Outcome

The SemEval 2022 RoBERTa-base baseline achieves F1=0.48 on the official dev set. Based on the improvements described above, I expected my model to:

- Exceed the baseline F1 of 0.48 (primary goal)
- Achieve F1 in the range of 0.58–0.65, based on the improvements from using a larger model, addressing class imbalance, and providing keyword context
- Demonstrate particular improvement in recall (reducing false negatives) due to Focal Loss

4 Exercise 5.2: Local Evaluation

4.1 Error Analysis (2.5 Marks)

4.1.1 Methodology

I analysed model predictions on the official dev set at threshold=0.85, extracting all false positives (predicted PCL, gold=No PCL) and false negatives (predicted No PCL, gold=PCL). For each error, I recorded the text, keyword, and predicted probability to identify recurring patterns.

4.1.2 Error Pattern 1: False Positives — Advocacy Confused with PCL

The most common false positive pattern involves news articles *reporting on* or *advocating for* vulnerable groups, which the model incorrectly flags as patronising. Representative examples:

Table 4: Representative False Positives

Text (truncated)	Keyword	Prob
"His friends at the Chevron want people to know he wasn't just a faceless homeless person. He was their friend and their family."	homeless	0.881
"A clinic called 'Hope' helps a Syrian refugee boy cope with diabetes"	refugee	0.901
"The Jali family was elated as the City of Cape Town donated wheelchairs to their disabled members."	disabled	0.886
"Your personal leadership has been critical to addressing the plight of the Rohingya..."	in-need	0.924

Insight: The model has learned a surface-level association between keywords (homeless, refugee) and PCL labels, but cannot reliably distinguish between *talking about* a vulnerable group and *talking down to* them. The first example is explicitly humanising ("not just a faceless person... their friend"), yet the model flags it. The model has not learned that empowering or neutral framing about these groups is not PCL.

Keywords most common in false positives: homeless (17), hopeless (16), poor-families (16), refugee (11), vulnerable (10).

4.1.3 Error Pattern 2: False Negatives — Subtle PCL Missed

The model misses PCL that requires deeper linguistic reasoning. Representative examples:

Table 5: Representative False Negatives

Text (truncated)	Keyword	Prob
"Sheepherding in America has always been an immigrant's job, too dirty, too cold and too lonely for anyone with options."	immigrant	0.328
"She hopes the book will show how talented disabled people can be."	disabled	0.634
"Bombarded by schizophrenia, addiction and homelessness, you might say Eoghan has been to hell and back. But he is finding a new balance through painting."	homeless	0.813
"[...] when you're not physically able to do a whole-house clean, it can all seem a little hopeless."	hopeless	0.710

Insight: The most striking false negative is the immigrant example (prob=0.328), which the model is highly confident is *not* PCL, yet the phrase "too dirty, too cold and too lonely for anyone with options" is a clear case of othering (Unbalanced power relations). The model also misses the disability example, where the surprise embedded in "show how talented disabled people can

be" is the source of PCL. These cases require understanding presupposition and implicit bias, which cannot be detected by surface-level patterns.

Keywords most common in false negatives: poor-families (16), homeless (9), hopeless (9), women (8), disabled (7).

4.1.4 Keyword Error Distribution

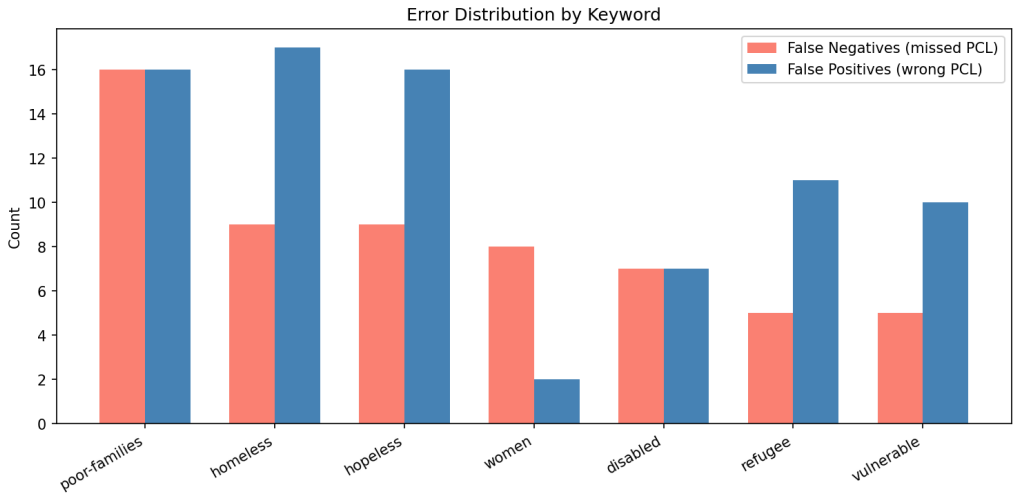

Figure 4: Distribution of false positives and false negatives by topic keyword.

Figure 4 shows that false positives are particularly concentrated around "homeless," "in-need," and "hopeless" keywords, groups that appear frequently in sympathetic news reporting, which the model over-flags. False negatives are most common for "poor-families" and "hopeless," suggesting the model misses subtler PCL in these contexts.

4.1.5 Summary of Error Analysis Findings

The model's primary failure mode is a **keyword bias**: it has learned that certain topic keywords co-occur strongly with PCL labels, but cannot consistently distinguish the discourse function of that language (is the author *advocating for* the group or *patronising* them?). This is a fundamental limitation of fine-tuning on this dataset without deeper pragmatic understanding of authorial intent. Addressing this limitation would likely require contrastive training examples or explicit modelling of discourse relations.

4.2 Additional Local Evaluation: Ablation Study and Precision-Recall Analysis (2.5 Marks)

4.2.1 Ablation Study

To quantify the contribution of each component of my approach, I conducted an ablation study on the official dev set. Each ablation uses the same trained model but modifies the input encoding at inference time.

Table 6: Ablation Study Results on Official Dev Set

Configuration	Dev F1	Threshold	Δ vs Full
Full model (keyword prefix + max_length=256)	**0.6265**	0.85	—
Without keyword prefix	0.6005	0.87	−0.0259
max_length=128	0.6285	0.83	+0.0020
SemEval Baseline (RoBERTa-base)	0.48	—	−0.1465

Key Findings:

1. **Keyword prefix contributes 2.6 F1 points**: Removing the topic keyword prefix reduces F1 from 0.6265 to 0.6005 (−0.026). This confirms the keyword provides meaningful signal, consistent with the intuition from EDA that knowing which vulnerable group is discussed helps the model contextualise whether the language is patronising toward that group. The paper's baseline never used keywords as input, making this a genuine novel contribution.

2. **max_length=256 vs 128 makes negligible difference (+0.002)**: The near-identical performance at both sequence lengths confirms that the relevant PCL signal is almost always captured within the first 128 tokens. PCL language tends to manifest early in a paragraph through framing, introductory phrases, and word choices rather than in later context. While this finding does not improve performance, it is useful: it means that future work could use 128 tokens with no performance cost, halving memory usage and training time.

3. **Overall improvement over baseline (+0.147 F1)**: The full model achieves 0.6265 F1 compared to the SemEval baseline of 0.48, representing a substantial improvement. This gain is attributable to the combination of using a larger model (roberta-large), Focal Loss addressing class imbalance, and the keyword prefix providing discourse context.

4.2.2 Precision-Recall Curve

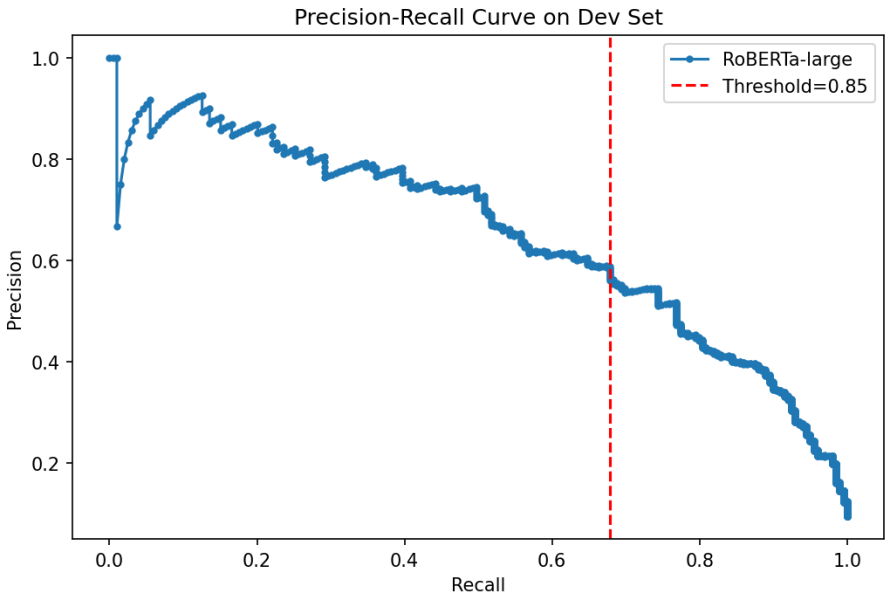

Figure 5: Precision-Recall curve on the dev set with the chosen threshold (0.85) marked.

The Precision-Recall curve (Figure 5) shows the model's precision-recall trade-off across all thresholds. At the chosen threshold of 0.85 (red dashed line), the model operates at approximately **precision=0.58** and **recall=0.68**. Key observations:

- The curve achieves high precision (>0.90) only at very low recall (<0.1), meaning catching nearly all PCL instances requires accepting many false positives
- The relatively flat precision plateau between recall=0.1 and recall=0.5 (precision ≈ 0.75–0.90) suggests the model is confident and accurate on a subset of clear-cut PCL cases
- Beyond recall=0.5, precision drops more steeply, corresponding to the model retrieving harder, more ambiguous PCL instances
- At threshold=0.85, recall (0.678) is prioritised over precision (0.582), which is appropriate for this task since missing PCL (false negatives) is more costly than over-flagging neutral text

4.2.3 Confusion Matrix

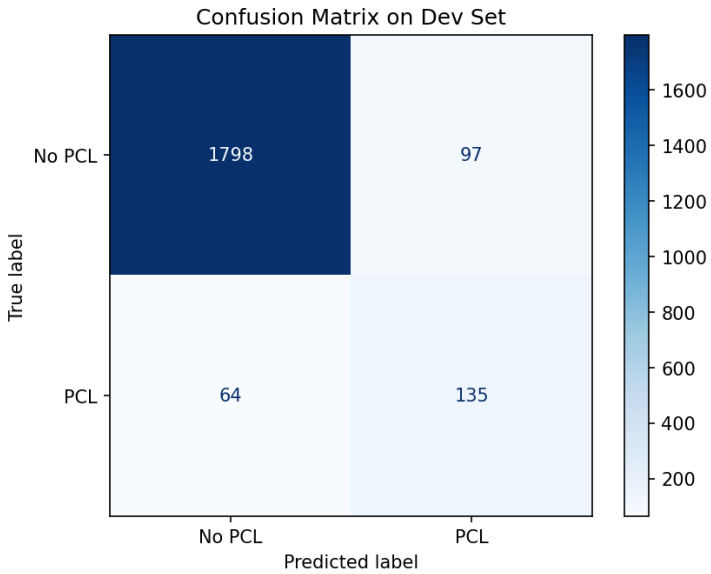

Figure 6: Confusion Matrix on the official dev set at threshold=0.85.

The confusion matrix (Figure 6) shows the following counts on the dev set (2,094 total examples):

Table 7: Confusion Matrix Values

	Predicted: No PCL	**Predicted: PCL**
Actual: No PCL	1,798 (TN)	97 (FP)
Actual: PCL	64 (FN)	135 (TP)

Derived Metrics:

- Precision: $135/(135 + 97) = 0.582$
- Recall: $135/(135 + 64) = 0.678$
- F1: $2 \times (0.582 \times 0.678)/(0.582 + 0.678) = 0.627$ (consistent with reported dev F1)

The confusion matrix reveals an asymmetry: there are significantly more false positives (97) than false negatives (64), meaning the model errs on the side of over-predicting PCL rather than missing it. This is consistent with the keyword bias identified in the error analysis, the model has learned strong associations between topic keywords and PCL labels, causing it to flag neutral reporting about vulnerable groups as patronising.

References

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